

TECHNICAL SPECIFICATION

# Ecclesia Digital Parish Manager (DPM)

System Features, Architecture Breakdown & End-to-End Test Cases Matrix

| VERSION                 | FRONTEND / STACK      | DATABASE                | AUTH & SECURITY              |
|-------------------------|-----------------------|-------------------------|------------------------------|
| v1.0 (Production-Ready) | Next.js 16 + React 19 | PostgreSQL + Prisma ORM | NextAuth v5 + 2FA / Passkeys |

## 1. System Architecture & High-Level Overview

Ecclesia is a multi-tenant Catholic parish management system that unifies administrative operations, sacramental records, financial stewardship, mass scheduling, pastoral appointments, and parishioner engagement into a single high-performance web platform.

KEY ARCHITECTURAL HIGHLIGHTS:

- Multi-Tenant Hierarchy: Two-level Parish to child Outstations scoping with data segregation.
- Step-Up Security Ladder: Multi-tier 2FA (TOTP / Recovery Keys), Passkeys (WebAuthn), and Re-Auth gates.
- Modular Feature Engine: 18+ per-organization configurable toggles in OrganizationFeatureSettings.
- Integrated Paystack Wallet: Subaccounts, Dedicated Virtual Bank Accounts, online & offline cash ledgering.

## 2. Core Functional Modules Breakdown

### 2.1 Multi-Tenant Organization & Hierarchy

- Hierarchical Scoping: Supports top-level Parishes and child Outstations linked via parentId.
- Tenant Scoping: All database access and actions are strictly scoped by session.user.organizationId.
- Inheritance & Roll-ups: Parish Admins view aggregate analytics across outstations; Outstation Admins stay isolated.

### 2.2 Role-Based Access Control (RBAC) & Security Ladder

- 8 User Roles: SUPER\_ADMIN, DIOCESE\_ADMIN, DEANERY\_ADMIN, PARISH\_ADMIN, PARISH\_SECRETARY, PARISH\_STAFF, SOCIETY\_PRESIDENT, PARISHIONER.
- Two-Factor Authentication (2FA): TOTP authenticator app support with encrypted single-use recovery codes.
- WebAuthn / Passkeys: Modern passwordless biometric login support across mobile and desktop browsers.
- Step-Up Re-Auth Gates: Critical mutations (updating bank payouts, role promotions) demand password/passkey re-auth.
- Parish Gate Code (/gate/[parishId]): Optional PIN gate to restrict public timelines for private parishes.

### 2.3 Parishioner Registry & Sacramental Tracking

- Parishioner Directory: Contact details, outstation affiliation, occupation tags, and family groupings.
- Batch CSV Import: High-throughput member onboarding with phone normalization (E.164) and validation.
- Sacramental Logbooks: Baptism, First Holy Communion, Confirmation, Marriage, and Holy Orders tracking.
- Certificate Generation: Automated PDF certificate generation with official parish headers and folios.

### 2.4 Liturgical Mass Management & Mass Intentions

- Recurring Mass Schedule Templates: Weekday morning, evening, and multiple Sunday Mass schedules.
- Automated Mass Generator: Auto-creates Mass instances up to 90 days in advance with capacity quotas.
- Mass Intention Booking: Online and offline booking for Thanksgiving, Repose of the Soul, and Special Intentions.
- Stipend Accounting & Quotas: Paystack card/USSD payments or manual cash marking with per-mass intention limits.
- Priest Mass Bulletin: Formatted printable sheet of booked intentions for the altar and liturgical ministers.

### 2.5 Financial Management & Paystack Wallet Integration

- Chart of Accounts: Tithes, Collections, Thanksgiving, Harvest, Building Projects, Mass Stipends, Society Dues.
- Paystack Subaccount & DVA: Automated split payouts, Dedicated Virtual Accounts (DVA) per parish, instant receipts.
- Cash Collections: Physical cash count entry with envelope numbers and counter sign-offs.
- Withdrawal Requests: Multi-step approval workflow to disburse parish wallet balances to verified commercial bank accounts.

### 2.6 Pastoral Appointments & Confessions

- Priest Availability Schedule: Recurring weekly availability slots and blackout dates (retreats, conferences).
- Booking Flow: Parishioners book pastoral direction, confessions, infant baptism prep, or general inquiries.
- Status Lifecycle: PENDING -> APPROVED / REJECTED -> COMPLETED / CANCELLED with instant notifications.

### 2.7 Societies & Pious Associations

- Society Directory: Sacred Heart, CWO, CMO, CYON, Legion of Mary, St. Vincent de Paul, Choir, etc.
- Executive Governance: Role assignments for President, Secretary, Financial Secretary, and Treasurer.
- Dues Tracking & Arrears: Automated monthly dues schedules, payment receipts, and arrears defaulter lists.

2.8 Public Parish Feed & Mobile Giving (/feed, /give, /me)

- Parish Timeline: Pinned announcements, saint reflections, liturgical calendar notices, and embedded live streams.
- Quick Giving (/give): Mobile-optimized giving flow via Card, USSD, or Bank Transfer.
- Member Portal (/me): Self-service giving receipts, booked mass intentions, appointments, and active sessions.

3. Comprehensive Test Cases & Verification Matrix

3.1 Authentication, RBAC & Security Ladder

| TEST ID    | SCENARIO                   | EXECUTION STEPS                                           | EXPECTED OUTCOME                                  | PRIORITY |
|------------|----------------------------|-----------------------------------------------------------|---------------------------------------------------|----------|
| TC-AUTH-01 | Standard Email Login       | 1. Open /auth/login<br>2. Submit valid credentials        | Session created; redirect to role dashboard/feed  | P0       |
| TC-AUTH-02 | TOTP 2FA Verification      | 1. Login on 2FA account<br>2. Enter valid 6-digit TOTP    | Session flag set to twoFactorVerified: true       | P0       |
| TC-AUTH-03 | 2FA Recovery Code Flow     | 1. Click 'Use recovery code'<br>2. Enter valid backup key | Code consumed; session unlocked with regen notice | P0       |
| TC-AUTH-04 | RBAC Boundary Isolation    | 1. Log in as PARISH_STAFF<br>2. Attempt admin API route   | HTTP 403 Forbidden / redirect to unauthorized     | P0       |
| TC-AUTH-05 | Sensitive Action Re-Auth   | 1. Navigate to Payout Settings<br>2. Update bank details  | Step-up modal requires password re-entry          | P1       |
| TC-AUTH-06 | Session Inactivity Timeout | 1. Simulate 24h idle<br>2. Request protected resource     | Session revoked; redirected to /auth/login        | P1       |
| TC-AUTH-07 | Parish Gate PIN Protection | 1. Visit gated /feed<br>2. Enter wrong & right PIN        | Wrong PIN rejects; correct PIN sets cookie        | P1       |

3.2 Organization & Mass Scheduling

| TEST ID    | SCENARIO                           | EXECUTION STEPS                                            | EXPECTED OUTCOME                                   | PRIORITY |
|------------|------------------------------------|------------------------------------------------------------|----------------------------------------------------|----------|
| TC-ORG-01  | Cross-Tenant Data Isolation        | 1. Create member in Parish A<br>2. Query as Parish B Admin | Parish A member is completely inaccessible to B    | P0       |
| TC-ORG-02  | Outstation Hierarchy Scope         | 1. View member list as Parish vs Outstation Admin          | Parish Admin sees aggregate; Outstation sees local | P0       |
| TC-MASS-01 | Automated Mass Generation          | 1. Configure recurring template<br>2. Run 30-day generator | 30 mass records created with accurate times & caps | P0       |
| TC-MASS-02 | Book Intention with Online Stipend | 1. Select Mass on calendar<br>2. Pay stipend via Paystack  | Webhook confirms charge; Intention CONFIRMED       | P0       |
| TC-MASS-03 | Intention Capacity Enforced        | 1. Attempt booking 6th intention on mass capped at 5       | Blocked: 'Mass Intention Capacity Full'            | P1       |
| TC-MASS-04 | Priest Mass Sheet Export           | 1. Open /masses for Sunday<br>2. Click 'Export Sheet'      | PDF bulletin generated grouping intentions by mass | P2       |

3.3 Financial Stewardship & Paystack

| TEST ID   | SCENARIO                  | EXECUTION STEPS                                         | EXPECTED OUTCOME                                   | PRIORITY |
|-----------|---------------------------|---------------------------------------------------------|----------------------------------------------------|----------|
| TC-FIN-01 | Record Offline Collection | 1. Open /parish-finances/new<br>2. Record Cash NGN 250k | Ledger updated; monthly analytics incremented      | P0       |
| TC-FIN-02 | Online Donation via /give | 1. Open /give on mobile<br>2. Pay NGN 10k via Card      | Paystack charge succeeds; ledger & receipt created | P0       |

|           |                              |                                                                |                                                  |    |
|-----------|------------------------------|----------------------------------------------------------------|--------------------------------------------------|----|
| TC-FIN-03 | Webhook Idempotency          | 1. Send duplicate webhook charge.success payload               | Recognized as duplicate; no double ledger credit | P0 |
| TC-FIN-04 | Withdrawal Approval Workflow | 1. Parish Admin requests withdrawal<br>2. Super Admin approves | Paystack Transfer API pays to verified bank      | P1 |

3.4 Sacraments, Appointments & Societies

| TEST ID   | SCENARIO                     | EXECUTION STEPS                                              | EXPECTED OUTCOME                                      | PRIORITY |
|-----------|------------------------------|--------------------------------------------------------------|-------------------------------------------------------|----------|
| TC-PAR-01 | Create Parishioner Record    | 1. Open /dashboard/parishioners/new<br>2. Fill info & submit | Record stored in DB scoped to current orgId           | P0       |
| TC-PAR-02 | Batch CSV Member Import      | 1. Upload 100-row CSV on /parishioners/import                | 100 records imported; validation errors logged        | P1       |
| TC-PAR-03 | Record Baptism & Certificate | 1. Fill Baptism details<br>2. Click 'Download PDF'           | Record saved; certificate downloaded with seal        | P1       |
| TC-APT-01 | Priest Appointment Booking   | 1. Pick Tuesday 10:00 AM slot<br>2. Submit purpose           | Appointment in PENDING state; staff notified          | P1       |
| TC-APT-02 | Approve Appointment Flow     | 1. Secretary clicks Approve on pending appointment           | Status set to APPROVED; member notified               | P1       |
| TC-SOC-01 | Society Executive Setup      | 1. Create Society 'CWO'<br>2. Designate President/Sec        | Society created; members receive society admin rights | P1       |
| TC-SOC-02 | Society Dues & Arrears       | 1. Set dues = NGN 1,000<br>2. Record member payments         | Paid members show 0 arrears; defaulters flagged       | P2       |

3.5 Non-Functional, Accessibility & Performance

| TEST ID   | SCENARIO              | EXECUTION STEPS                                     | EXPECTED OUTCOME                                | PRIORITY |
|-----------|-----------------------|-----------------------------------------------------|-------------------------------------------------|----------|
| TC-NFR-01 | Accessibility (a11y)  | Run automated Playwright @axe-core/playwright audit | 0 critical axe-core violations (WCAG 2.1 AA)    | PASS     |
| TC-NFR-02 | Mobile Responsiveness | Test fluid layout across 375px - 1920px viewports   | Zero layout breaks or horizontal overflow       | PASS     |
| TC-NFR-03 | API Rate Limiting     | Simulate burst brute-force login and API requests   | 5 req/min on login, 60 req/min on standard APIs | PASS     |
| TC-NFR-04 | Core Web Vitals       | Lighthouse performance run on /feed and /give       | LCP < 2.0s, TTFB < 400ms on broadband           | PASS     |

4. Automated Test Execution Commands

```
# Run all Playwright E2E test suites
pnpm exec playwright test

# Run critical security & authentication test specs
pnpm exec playwright test e2e/p0-2fa-login-bug.spec.ts e2e/p1-security-ladder.spec.ts

# Run TypeScript type checks and linting
pnpm tsc --noEmit && pnpm lint
```